

Prefix-Outcome Validation for On-Policy Distillation of Reasoning Models

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Abstract

On-policy distillation (OPD) trains a student on its own sampled trajectories using dense token-level preferences from a stronger teacher. This provides fine-grained supervision on student-visited states, but it also assumes that every teacher logit difference is reliable along the whole sampled trajectory. In long-horizon reasoning, this assumption can fail: a local teacher preference may conflict with the final correctness of the response, and later prefixes may drift into regions where teacher guidance is less trustworthy. We propose *Prefix-Outcome Validation* (POV), a validation layer for OPD that calibrates dense teacher supervision using two complementary signals. Outcome validation uses final correctness to determine whether each teacher-student logit difference is aligned with the trajectory result, preserving aligned candidates while assigning only beta-capped rank weights to anti-aligned candidates. Prefix validation monitors teacher excess surprisal along the generated prefix and monotonically decays token supervision after sustained prefix drift. POV preserves OPD’s dense supervision while grounding it in both final outcome consistency and prefix-level teacher support, yielding a conservative alternative to outcome-only RL and unvalidated dense distillation.

Introduction

Outcome-based reinforcement learning has become a standard post-training tool for improving the reasoning ability of language models. In mathematical reasoning, a verifier can often judge whether a completed response is correct, and policy optimization can then increase the probability of successful responses while decreasing the probability of failed ones (Schulman et al. 2017; Ouyang et al. 2022; Shao et al. 2024; DeepSeek-AI 2025). Group-relative methods such as GRPO make this setup especially simple: several responses are sampled for the same prompt, a scalar outcome reward is computed for each response, and the resulting relative advantage is broadcast to all response tokens. This design uses reliable final feedback, but it provides little information about which intermediate tokens should be reinforced.

On-policy distillation (OPD) offers a complementary source of supervision. Instead of training only from a scalar outcome, the student first samples its own response and a stronger teacher is then evaluated on the prefixes actually visited by the student. The teacher provides dense token-level guidance, often through the log-probability gap

between the teacher and the student. This keeps training on-policy while exposing the student to fine-grained teacher preferences. For long reasoning traces, however, dense teacher supervision is not uniformly reliable. The teacher may locally prefer a token that belongs to an ultimately incorrect solution, or locally disfavor a token that appears in a correct trajectory. This mismatch creates a credit-assignment problem: the local teacher signal and the final outcome do not always agree.

A second issue arises from the prefixes on which the teacher is queried. OPD evaluates the teacher on student-generated prefixes rather than on teacher-generated ones. Once a student trajectory drifts into a low-support region under the teacher, later teacher logit differences still exist, but they may be less informative about how to reach a correct solution. Treating these later dense rewards as uniformly trustworthy can amplify supervision from invalid or off-path prefixes. Thus, OPD needs not only token-outcome consistency, but also a way to judge whether the current prefix remains a valid context for teacher supervision.

We propose *Prefix-Outcome Validation* (POV), a simple validation mechanism for on-policy distillation. POV uses final correctness to validate the direction of dense teacher preferences, rather than replacing them with a sparse reward. In a correct trajectory, a candidate token is outcome-aligned when the teacher assigns it higher log-probability than the student. In an incorrect trajectory, the alignment direction is reversed: a candidate is outcome-aligned when the teacher assigns it lower log-probability than the student, thereby discouraging behavior observed in a failed response. This rule converts the final verifier result into a token-level reliability test for OPD supervision.

POV also validates prefixes. Since the teacher is queried on student-generated prefixes, the later parts of a response may drift into regions where the teacher’s dense preferences are less trustworthy. POV monitors teacher excess surprisal, defined as the negative teacher log-probability of the sampled token minus the teacher entropy at the same prefix. A sustained increase in excess surprisal indicates that the sampled prefix has become unexpectedly unlikely under the teacher. POV detects this change with a windowed CUSUM statistic and monotonically decays subsequent dense supervision after the detected horizon.

The resulting method, POV-OPD, keeps the useful part

of dense OPD while reducing the influence of locally unreliable signals. Outcome validation assigns full weight to outcome-aligned token candidates and uses a beta-capped rank weight for anti-aligned candidates. Prefix validation then attenuates supervision after sustained teacher-surprisal drift. Thus, POV-OPD preserves token-level teacher guidance when it is consistent with both the final outcome and the teacher’s prefix-level support.

Our contributions are:

- We identify two reliability issues in on-policy distillation for reasoning: outcome-token mismatch and prefix drift under student-generated trajectories.
- We introduce outcome validation, which uses final correctness to partition dense teacher preferences into outcome-aligned and anti-aligned token candidates.
- We introduce prefix validation, which detects sustained excess surprisal under the teacher and monotonically down-weights later dense supervision.
- We combine both mechanisms into POV-OPD, a conservative modification of dense OPD that preserves aligned supervision and assigns only beta-capped rank weight to anti-aligned candidates.

Related Work

Outcome-Based Reinforcement Learning for Reasoning

Reinforcement learning from outcome feedback is widely used to align language models and improve their reasoning behavior. PPO-style RLHF optimizes a policy against learned or programmatic rewards while controlling policy drift (Christiano et al. 2017; Schulman et al. 2017; Ouyang et al. 2022). Direct preference optimization and related preference-learning methods simplify parts of this pipeline when pairwise preference data are available (Rafailov et al. 2023). For mathematical reasoning, recent systems often use verifiable rewards, where a final answer can be checked automatically (Shao et al. 2024; DeepSeek-AI 2025). GRPO and related group-relative methods reduce the need for a learned value function by comparing multiple responses sampled for the same prompt and broadcasting a scalar relative advantage over response tokens (Shao et al. 2024). These methods use reliable final feedback, but they do not provide fine-grained token-level guidance. POV-OPD addresses this limitation by retaining dense teacher supervision while using final correctness as a validation signal.

Knowledge Distillation and On-Policy Distillation

Knowledge distillation transfers behavior from a stronger teacher model to a smaller or weaker student (Hinton, Vinyals, and Dean 2015). For sequence generation, distillation can imitate teacher outputs or teacher distributions at the token or sequence level. Offline distillation is simple, but it trains the student on states induced by the teacher or by a fixed dataset, which can differ from the states visited by the student during reasoning. On-policy distillation reduces this mismatch by evaluating teacher preferences on student-generated trajectories. Recent work on OPD shows

that dense token-level teacher rewards can be useful but sensitive to teacher-student compatibility and the states visited by the student (Li et al. 2026). POV-OPD follows the same on-policy view: the student samples the response, the teacher scores the visited prefixes, and the student updates from dense teacher-student log-probability gaps. The key difference is that POV-OPD does not assume every dense teacher preference is equally reliable.

Credit Assignment and Dense Supervision

Long-horizon reasoning requires assigning credit to intermediate decisions from sparse final outcomes. Chain-of-thought prompting, self-consistency, and self-training methods highlight the importance of intermediate reasoning paths for final-answer accuracy (Wei et al. 2022; Wang et al. 2022; Zelikman et al. 2022). Process supervision provides dense feedback for intermediate steps, but such supervision can be costly or noisy (Lightman et al. 2023). Dense teacher rewards offer an automatic alternative, yet they can conflict with the final correctness of the trajectory. POV-OPD treats this conflict as a validation problem: final correctness determines whether the local teacher-student preference points in an outcome-consistent direction. Our prefix validation is also related to classical change detection, where CUSUM statistics are used to detect sustained shifts in a monitored signal (Page 1954; Basseville and Nikiforov 1993). Here the monitored signal is teacher excess surprisal along a student-generated prefix, and the response is a monotone decay of later OPD weights rather than a hard rejection of the trajectory.

Method

Problem Setup

Let x be a prompt, π_θ be the student policy, and π_T be a fixed teacher policy. The student samples a response $y_i = (y_{i,1}, \dots, y_{i,T_i}) \sim \pi_\theta(\cdot | x)$. For each response position t , we consider a candidate set $\mathcal{C}_{i,t} = \{a_{i,t,k}\}_{k=1}^K$. In the sampled-token setting, $K = 1$ and $a_{i,t,1} = y_{i,t}$. In the top- K setting, candidates may come from the student distribution, the teacher distribution, or their union.

On-policy distillation computes teacher and student log-probabilities on the same student-visited prefix:

$$\begin{aligned}\ell_{i,t,k}^T &= \log \pi_T(a_{i,t,k} | x, y_{i,<t}), \\ \ell_{i,t,k}^\theta &= \log \pi_\theta(a_{i,t,k} | x, y_{i,<t}).\end{aligned}\tag{1}$$

The basic teacher-student log-probability gap is

$$\delta_{i,t,k} = \ell_{i,t,k}^T - \ell_{i,t,k}^\theta.\tag{2}$$

The dense OPD advantage can be written as

$$D_{i,t,k} = \rho_{i,t,k} \delta_{i,t,k},\tag{3}$$

where $\rho_{i,t,k}$ is an optional candidate weight. For sampled-token OPD, $\rho_{i,t,k} = 1$. Standard dense OPD directly uses

$$A_{i,t,k}^{\text{OPD}} = D_{i,t,k}.\tag{4}$$

Each sampled response also receives an outcome score R_i , provided by the task verifier. In our implementation this

is the field `true_reward_score`. We convert it to a binary correctness label

$$c_i = \mathbf{1}[R_i > 1/2], \quad (5)$$

and a signed outcome label

$$z_i = 2c_i - 1. \quad (6)$$

Thus $z_i = 1$ for a correct response and $z_i = -1$ for an incorrect response.

GRPO, OPD, and POV-OPD

Figure 1 summarizes the difference between outcome-only RL, dense OPD, and POV-OPD. GRPO uses only the final outcome. Given a group of responses sampled for the same prompt, it computes a relative outcome advantage, for example

$$\hat{R}_i = \frac{R_i - \mu(x_i)}{\sigma(x_i) + \epsilon}, \quad (7)$$

where $\mu(x_i)$ and $\sigma(x_i)$ are computed over responses to the same prompt. This scalar advantage is then broadcast to all response tokens:

$$A_{i,t}^{\text{GRPO}} = \hat{R}_i. \quad (8)$$

GRPO therefore gives reliable trajectory-level feedback, but it does not distinguish which local token decisions were useful.

OPD takes the opposite approach. It provides dense token-level teacher guidance on the student trajectory, but it does not check whether a local teacher preference is consistent with the final outcome. POV-OPD combines the two views. It keeps the dense OPD advantage $D_{i,t,k}$, but multiplies it by validation weights derived from the final outcome and from teacher prefix statistics:

$$A_{i,t,k}^{\text{POV}} = D_{i,t,k} \cdot w_{i,t,k}^{\text{out}} \cdot w_{i,t}^{\text{pre}} \cdot b_{i,t,k}. \quad (9)$$

Here w^{out} is the outcome-validation weight, w^{pre} is the prefix-validation weight, and b is an aligned-candidate boost.

Outcome Validation

Outcome validation checks whether the teacher-student preference points in the direction suggested by the final outcome. We use the alignment score

$$g_{i,t,k} = \delta_{i,t,k}, \quad (10)$$

and define the signed alignment score

$$s_{i,t,k} = z_i g_{i,t,k}. \quad (11)$$

For a correct response, $z_i = 1$, so a candidate is aligned when the teacher assigns it higher log-probability than the student. For an incorrect response, $z_i = -1$, so a candidate is aligned when the teacher assigns it lower log-probability than the student. With margin $m \geq 0$, the aligned and anti-aligned sets are

$$\mathcal{A}_i = \{(t, k) : s_{i,t,k} > m\}, \quad \mathcal{B}_i = \{(t, k) : s_{i,t,k} < -m\}. \quad (12)$$

Candidates with $|s_{i,t,k}| \leq m$ are treated as ambiguous.

The hard outcome-validation weight keeps only aligned candidates:

$$w_{i,t,k}^{\text{out}} = \mathbf{1}[(t, k) \in \mathcal{A}_i]. \quad (13)$$

This removes outcome-conflicting dense rewards, but it can be brittle when either the local teacher preference or the final verifier signal is noisy.

POV-OPD therefore also uses a rank variant. The rank variant preserves all aligned candidates at full weight and assigns only beta-capped weights to anti-aligned candidates. Let $n_i = |\mathcal{B}_i|$. For $(t, k) \in \mathcal{B}_i$, define

$$q_{i,t,k} = \begin{cases} 1, & n_i = 1, \\ \frac{\text{rank}_{\mathcal{B}_i}(s_{i,t,k})}{n_i - 1}, & n_i > 1. \end{cases} \quad (14)$$

The lowest signed score in $\mathcal{B}_i$ has rank 0, and the largest signed score in $\mathcal{B}_i$ has rank $n_i - 1$. Thus anti-aligned candidates closer to the margin receive larger rank weights. With $\beta \in [0, 1]$, the rank outcome-validation weight is

$$w_{i,t,k}^{\text{out}} = \begin{cases} 1, & (t, k) \in \mathcal{A}_i, \\ \beta q_{i,t,k}, & (t, k) \in \mathcal{B}_i, \\ 0, & |s_{i,t,k}| \leq m. \end{cases} \quad (15)$$

This differs from globally reweighting all candidates: aligned candidates are not down-weighted, and β only controls the maximum retained strength of anti-aligned supervision.

POV-OPD also supports position-level validation. Instead of validating each candidate separately, it first aggregates the alignment score at a position:

$$s_{i,t}^{\text{pos}} = z_i \sum_{k=1}^K g_{i,t,k}. \quad (16)$$

The hard position variant keeps positions with $s_{i,t}^{\text{pos}} > m$. The rank position variant ranks only positions with $s_{i,t}^{\text{pos}} < -m$, assigns them beta-capped weights, and broadcasts the resulting position weight to all candidates at that position. These variants test whether outcome consistency should be judged at the candidate level or at the next-token decision level.

Prefix Validation

Outcome validation checks the direction of local teacher preferences. Prefix validation checks whether the teacher remains reliable on the student-generated prefix. Let

$$H_{i,t}^T = - \sum_a \pi_T(a \mid x, y_{i,<t}) \log \pi_T(a \mid x, y_{i,<t}) \quad (17)$$

be the teacher entropy at position t . For the sampled token $y_{i,t}$, define teacher excess surprisal

$$u_{i,t} = - \log \pi_T(y_{i,t} \mid x, y_{i,<t}) - H_{i,t}^T. \quad (18)$$

A positive value means that the sampled token is more surprising under the teacher than a typical token drawn from the teacher distribution at the same prefix.

For each response, we divide valid response tokens into consecutive windows of size W . Let $\bar{u}_{i,b}$ be the mean excess

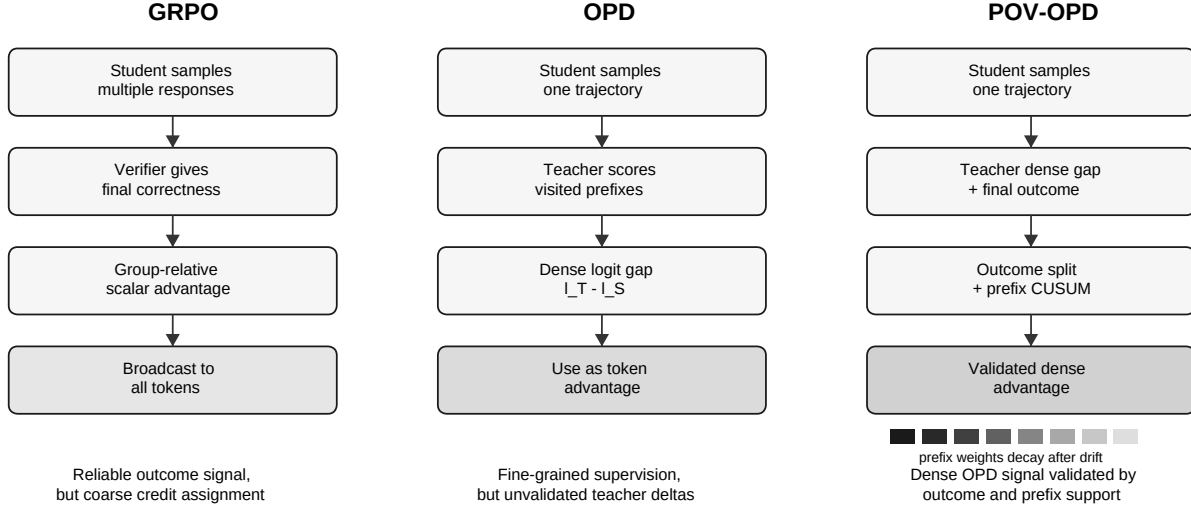


Figure 1: Comparison of outcome-only RL, dense OPD, and POV-OPD. GRPO uses final correctness to form a group-relative scalar advantage and broadcasts it to all response tokens. OPD queries the teacher on student-visited prefixes and uses dense teacher-student log-probability gaps directly as token-level supervision. POV-OPD keeps dense OPD supervision but validates it with two signals: outcome validation preserves outcome-aligned token candidates and assigns only beta-capped rank weight to anti-aligned candidates, while prefix validation monitors teacher excess surprisal and decays later supervision after sustained prefix drift.

surprisal in window b . The first L windows form a baseline, with

$$\mu_i = \frac{1}{L_i} \sum_{b=1}^{L_i} \bar{u}_{i,b}, \quad L_i = \min(L, M_i), \quad (19)$$

where M_i is the number of windows in response i . For later windows, we compute relative excess

$$d_{i,b} = \bar{u}_{i,b} - \mu_i, \quad (20)$$

and a one-sided CUSUM statistic

$$C_{i,b} = \max(0, C_{i,b-1} + d_{i,b} - \kappa), \quad b > L_i, \quad (21)$$

where $\kappa \geq 0$ is a drift allowance. The prefix horizon is the first post-baseline window whose CUSUM exceeds threshold τ :

$$b_i^* = \min\{b > L_i : C_{i,b} > \tau\}. \quad (22)$$

If no such window exists, all prefix weights remain one.

After the horizon is triggered, POV-OPD monotonically decays the prefix weight:

$$\eta_{i,b} = \exp[-\lambda \max(C_{i,b} - \tau, 0)],$$

$$w_{i,b}^{\text{pre}} = \begin{cases} 1, & b < b_i^*, \\ \min(w_{i,b-1}^{\text{pre}}, \eta_{i,b}), & b \geq b_i^*. \end{cases} \quad (23)$$

The token-level prefix weight $w_{i,t}^{\text{pre}}$ is obtained by assigning the window weight to every valid token in that window. This produces a non-increasing trust weight after sustained teacher-surprisal drift.

Aligned-Candidate Support

POV-OPD gives a small boost to aligned candidates that are also well supported by the teacher distribution. For a candidate $a_{i,t,k}$, define

$$r_{i,t,k} = -\ell_{i,t,k}^T - H_{i,t}^T. \quad (24)$$

The candidate support is

$$p_{i,t,k}^{\text{sup}} = \exp[-\max(r_{i,t,k}, 0)]. \quad (25)$$

This support equals one when the candidate is no more surprising than the teacher entropy and decays when the candidate is unusually surprising under the teacher. The aligned boost is

$$b_{i,t,k} = 1 + \alpha p_{i,t,k}^{\text{sup}} \mathbf{1}[(t, k) \in \mathcal{A}_i], \quad (26)$$

where $\alpha \geq 0$ controls the boost strength. Anti-aligned and ambiguous candidates are not boosted.

Final Objective

The final POV-OPD advantage is

$$A_{i,t,k}^{\text{POV}} = D_{i,t,k} \cdot w_{i,t,k}^{\text{out}} \cdot w_{i,t}^{\text{pre}} \cdot b_{i,t,k}. \quad (27)$$

For sampled-token OPD, this expression reduces to $K = 1$. For top- K OPD, it applies to each candidate token.

The policy is optimized with the same PPO-style update used by OPD. Let

$$r_{i,t,k}(\theta) = \frac{\pi_{\theta}(a_{i,t,k} \mid x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(a_{i,t,k} \mid x, y_{i,<t})}. \quad (28)$$

Let $\bar{r}_{i,t,k}(\theta) = \text{clip}(r_{i,t,k}(\theta), 1 - \epsilon, 1 + \epsilon)$. The clipped objective is

$$\mathcal{L}_{\text{POV}} = - \sum_{i,t,k} \min(r_{i,t,k}(\theta)A_{i,t,k}^{\text{POV}}, \bar{r}_{i,t,k}(\theta)A_{i,t,k}^{\text{POV}}). \quad (29)$$

POV-OPD therefore changes the credit assignment used to form advantages, while leaving rollout generation, teacher scoring, and the actor optimization pipeline unchanged.

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